

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Objective

To understand Azure cloud concepts and implement an end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The project demonstrates cloud resource creation, storage management, metadata validation, data movement, pipeline execution, and access control management.

Task 1: Explore Azure Portal and Create a Resource Group

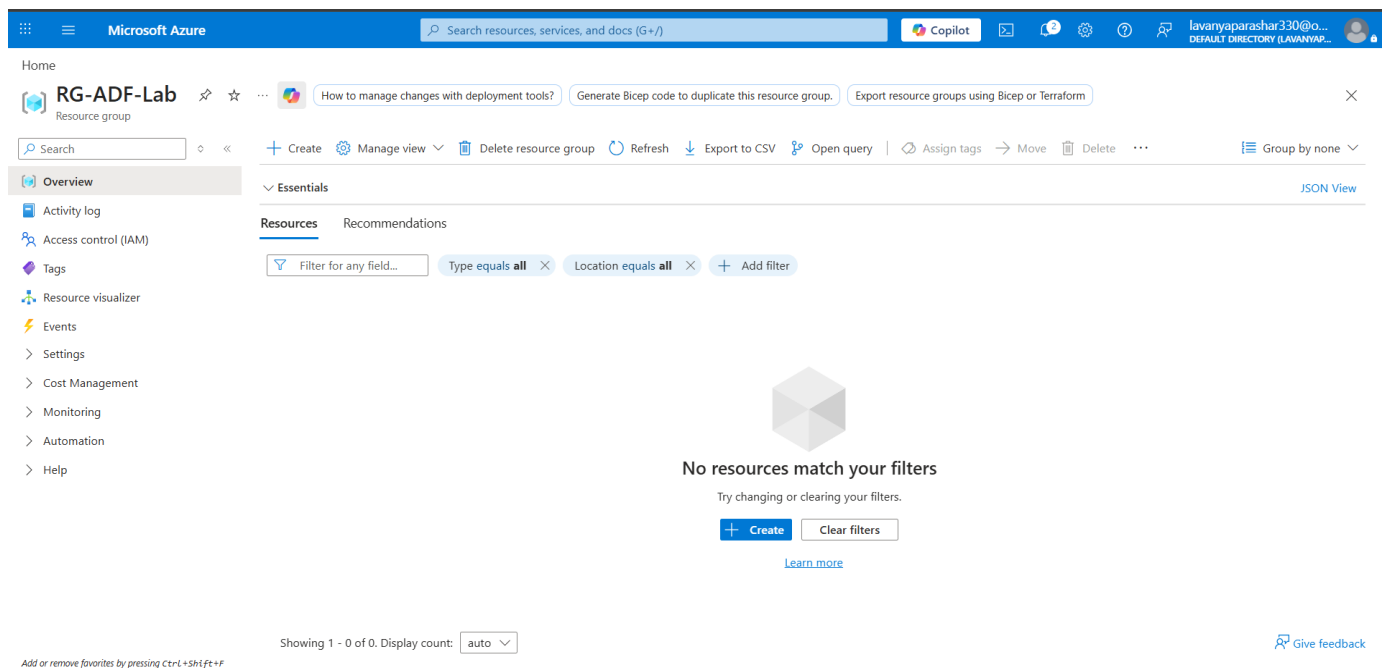
Description

A Resource Group was created in Azure Portal to logically organize and manage all Azure resources required for the project. The Resource Group acts as a container that holds related Azure services such as Storage Account and Azure Data Factory.

Steps Performed

1. Logged in to Azure Portal.
2. Navigated to Resource Groups.
3. Created a new Resource Group.
4. Selected the appropriate region.
5. Successfully deployed the Resource Group.
- 6.

Screenshot 1



Resource Group Overview Page

Task 2: Storage Setup

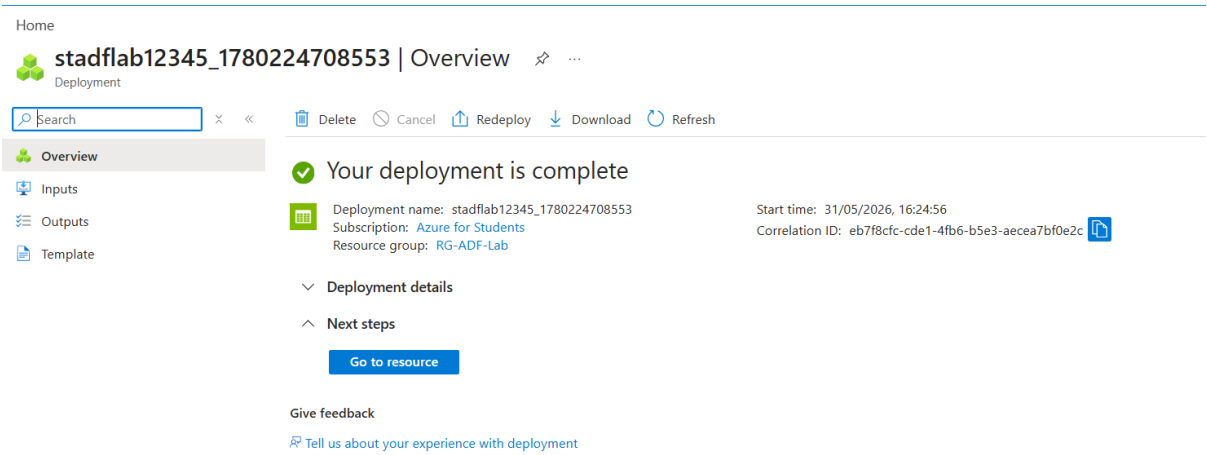
Description

An Azure Storage Account was created to store the dataset used in the project. Blob Storage was used because it is optimized for storing files such as CSV datasets.

Steps Performed

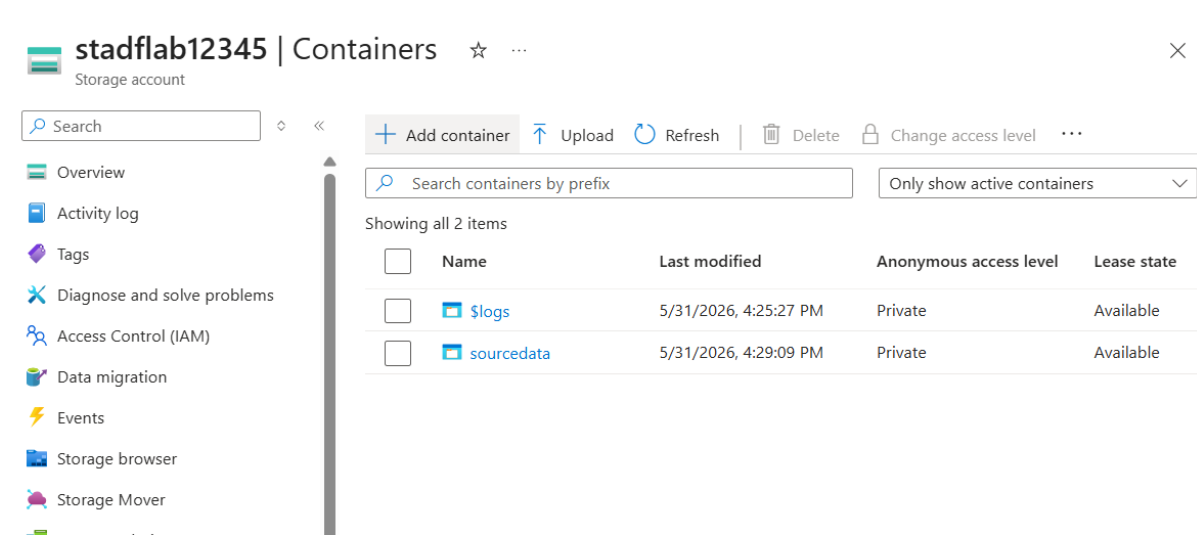
1. Created a Storage Account.
2. Configured standard performance settings.
3. Selected Locally Redundant Storage (LRS).
4. Created a Blob Container named sourcedata.
5. Uploaded the Superstore.csv dataset into the container.

Screenshot 2



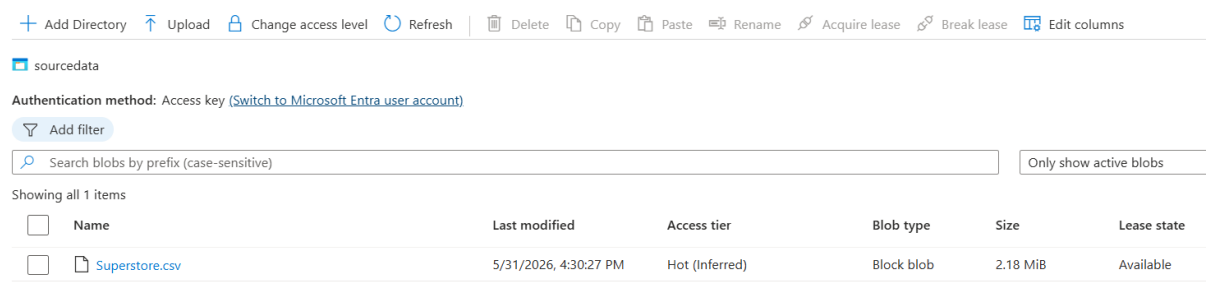
Storage Account Overview Page

Screenshot 3



sourcedata Container Created

Screenshot 4



sourcecontainer Showing Superstore.csv

Task 3: Azure Data Factory Basics

Description

Azure Data Factory was created to build and manage the data pipeline. A Linked Service was configured to establish connectivity between Azure Data Factory and Azure Blob Storage.

Datasets were created to represent the source and destination files used in the pipeline.

Components Created

Linked Service

Name: LS_BlobStorage

Purpose:

To connect Azure Data Factory with Azure Blob Storage.

Source Dataset

Name: DS_Source

Purpose:

To access Superstore.csv stored in Blob Storage.

Destination Dataset

Name: DS_Destination

Purpose:

To store the copied file in the processed container.

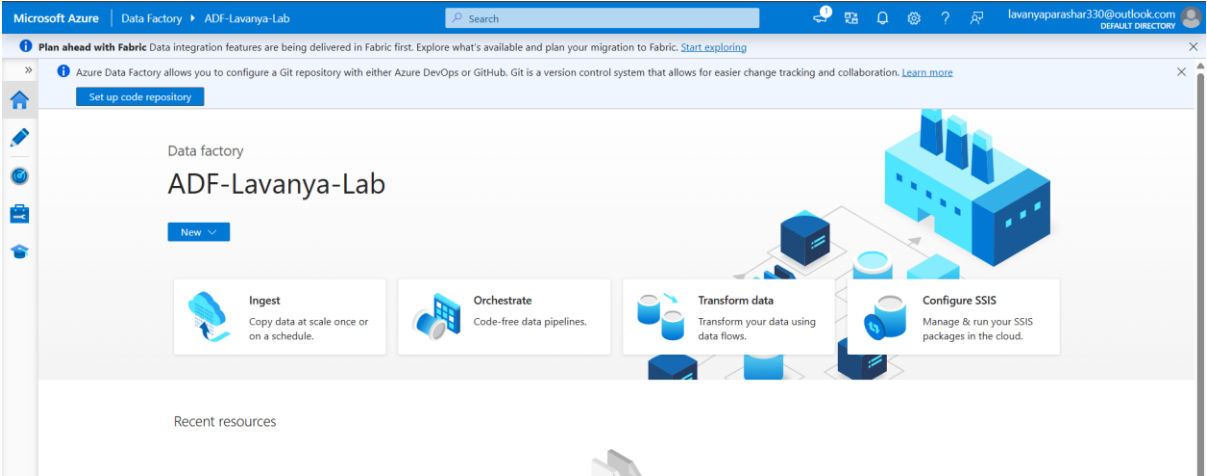
Get Metadata Activity

The Get Metadata activity was configured to retrieve:

- Exists
- Size
- Last Modified

This activity validates file information before processing.

Screenshot 5



Azure Data Factory Setup Screen

Screenshot 6

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

[+ New](#)

Annotations : **Any**

Showing 1 - 1 of 1 items

Name ↑↓	Type ↑↓	Related ↑↓	Annotations ↑↓
LS_BlobStorage	Azure Blob Storage	0	

Linked Service Screen

Screenshot 7

stadflab12345 | Containers ☆ ...

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

+ Add container

↑ Upload

↻ Refresh

🗑 Delete

🔒 Change access level

⋮

Search containers by prefix

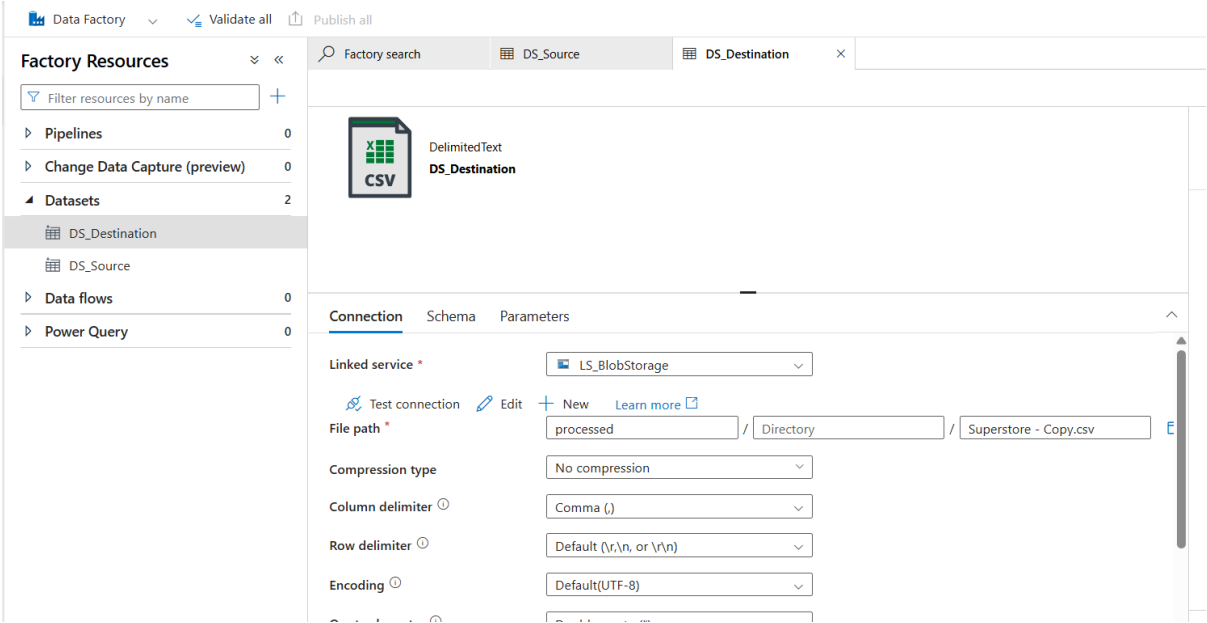
Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state
☐	\$logs	5/31/2026, 4:25:27 PM	Private	Available
☐	processed	5/31/2026, 5:07:11 PM	Private	Available
☐	sourcedata	5/31/2026, 4:29:09 PM	Private	Available

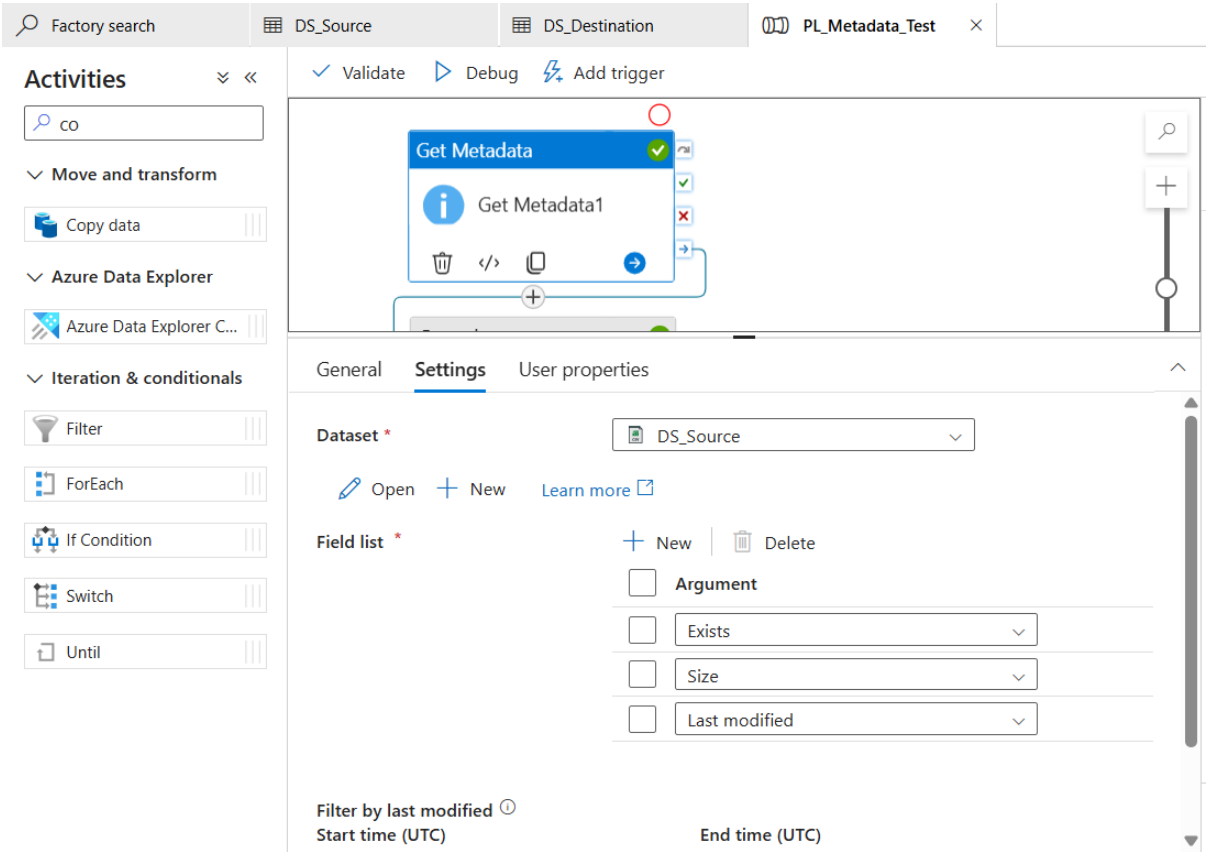
processed Container Created

Screenshot 8



Published Datasets Screen

Screenshot 9



Metadata Activity Fields (Exists, Size, Last Modified)

Task 4: Pipeline Development

Description

A pipeline was created using Get Metadata and Copy Data activities.

The pipeline first validates the source file metadata and then copies the file from the source container to the destination container.

Pipeline Flow

Source File (Superstore.csv)



Get Metadata

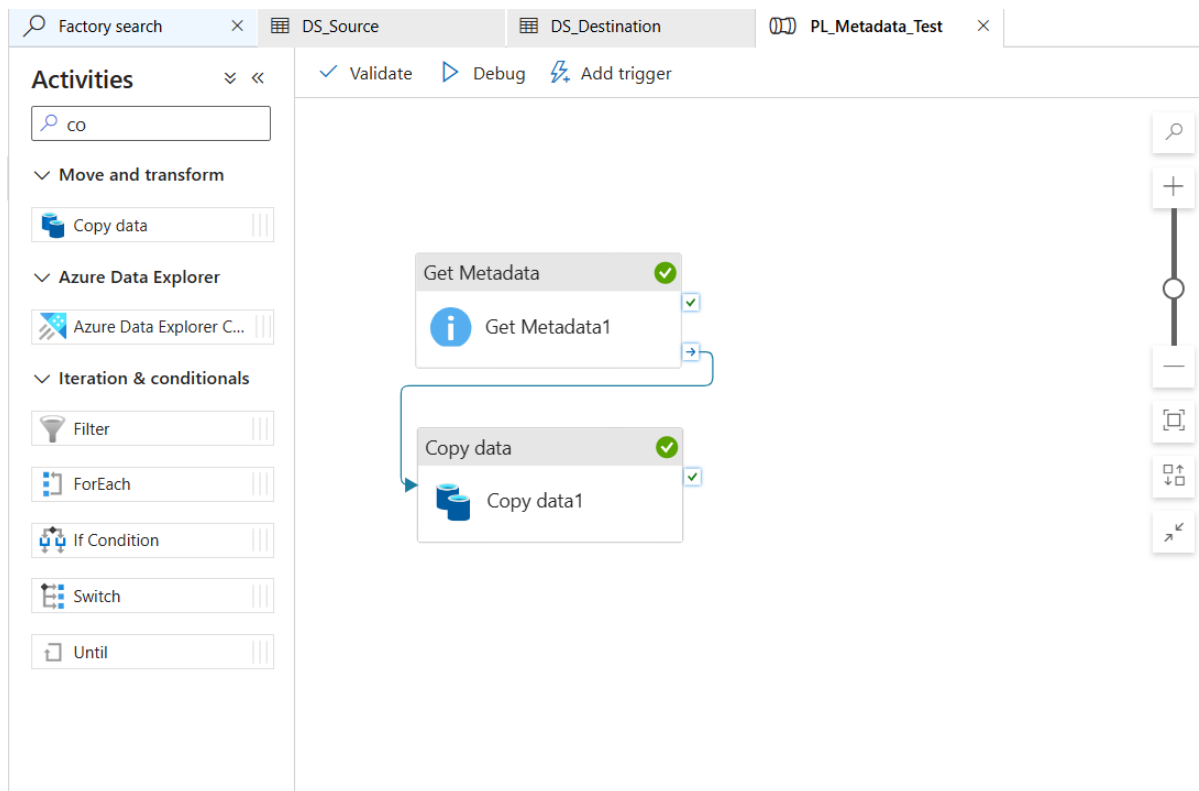


Copy Data



Destination File (Superstore_Copy.csv)

Screenshot 10



Pipeline Canvas Screenshot

Task 5: Pipeline Execution

Description

The pipeline was executed using the Debug feature in Azure Data Factory.

Execution results showed that both activities completed successfully.

Activities Executed

1. Get Metadata – Succeeded
2. Copy Data – Succeeded

Outcome

The source file was successfully copied to the destination container.

Screenshot 11

Parameters

Variables

Settings

Output

Pipeline

f1e8b87b-de0e-4ed1-a

run ID

f01-b8599dd741f1

Pipeline status

✔ Succeeded

[View debug run consumption](#)

All status

[Monitor in Azure Metrics](#)

[Export to CSV](#)

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Durat
Copy data1	✔ Succeeded	Copy data	5/31/2026, 5:23:43 PM	16s
Get Metadata1	✔ Succeeded	Get Metadata	5/31/2026, 5:23:26 PM	16s

Pipeline Successfully Working Screenshot

Task 6: IAM Role Assignment

Description

Azure Identity and Access Management (IAM) was explored to understand access control mechanisms.

Role assignment options were reviewed and Reader access was configured. Contributor role options were also explored.

Roles Explored

- Reader
- Contributor
- Specialized Contributor Roles

Purpose:
To understand how Azure controls permissions and resource access.

Screenshot 12

Home > Storage center | Blob Storage > stadflab12345 | Access Control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Copilot can help pick a role

Job function roles

Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

contributor

Type : All

Category : All

Name	Description	Type	Category	Details
Advisor Reviews Contributor	View reviews for a workload and triage recommendations linked to them.	BuiltInRole	None	View
Anyscale Platform Contributor Role	Grants read/write access to Anyscale Cloud and resources	BuiltInRole	None	View
API Management Service Contributor	Can manage service and the APIs	BuiltInRole	Integration	View
API Management Workspace Contributor	Can manage the workspace and view, but not modify its members. This role should be assigned on the workspace scope.	BuiltInRole	None	View
App Configuration Contributor	Grants permission for all management operations, except purge, for App Configuration resources. This role does not grant...	BuiltInRole	None	View
App Service Environment Contributor	Manage App Service Environments but not the App Service Plans or Websites that it hosts.	BuiltInRole	None	View

Contributor Role Assignment Options

Screenshot 13

role-assignments-2026-05-31.csv

role-assignments-2026-05-31.json

C: > Users > lavan > Downloads > role-assignments-2026-05-31.json > 1

```
1 [{"RoleAssignmentId": "76f14a4a-4be9-48ad-9473-f37c33c8626a",
2  "Scope": "/subscriptions/b2fefccd-c6df-43da-a458-dcead1084e23",
3  "DisplayName": "Lavanya Parashar",
4  "SignInName": "lavanyaparashar330_outlook.com#EXT#@lavanyaparashar330outlook.onmicrosoft.com",
5  "RoleDefinitionName": "Owner",
6  "RoleDefinitionId": "/subscriptions/b2fefccd-c6df-43da-a458-dcead1084e23/providers/Microsoft.Autho
7  "ObjectId": "0ad394e4-aa50-47ff-9ae2-11c0a9b7eb06",
8  "ObjectType": "User",
9  "RoleAssignmentDescription": "",
10 "ConditionVersion": "",
11 "Condition": ""},
```

Saved Role Assignment JSON Screenshot

Mini Project: End-to-End Data Pipeline

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory.

Architecture

Blob Storage (Superstore.csv)

↓

Linked Service

↓

Dataset (Source)

↓

Get Metadata

↓

Copy Data

↓

Dataset (Destination)

↓

Processed Container

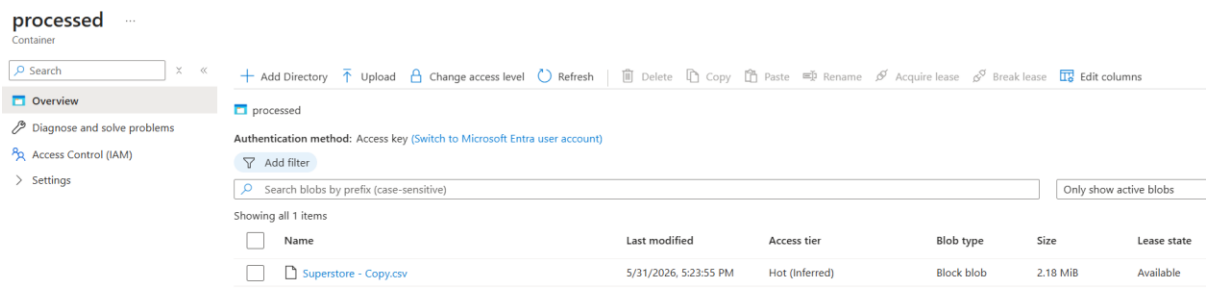
Implementation

1. Uploaded Superstore.csv to Blob Storage.
2. Connected Azure Data Factory using Linked Service.
3. Created source and destination datasets.
4. Configured Get Metadata activity.
5. Configured Copy Data activity.
6. Executed the pipeline.
7. Verified output in the processed container.

Results

- Metadata validation completed successfully.
- File existence was verified.
- File size and last modified information were retrieved.
- Data was copied successfully.
- Output file was generated in the destination location.

Screenshot 14



Processed Container Showing Superstore_Copy.csv

Conclusion

This project successfully demonstrated the implementation of an end-to-end Azure Data Pipeline using Azure Storage Account and Azure Data Factory.

The pipeline validated metadata using Get Metadata activity and copied data from the source container to the destination container using Copy Data activity. Pipeline execution completed successfully, confirming proper integration between Azure Blob Storage and Azure Data Factory.

The project provided practical experience with Azure cloud services, storage management, pipeline orchestration, metadata validation, data movement, and access control management.